

μC



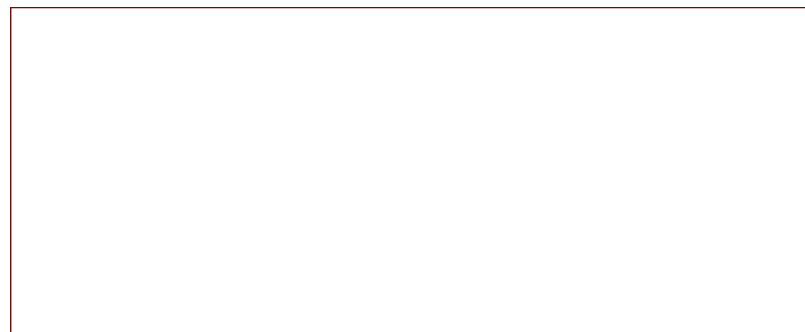
File: μC.kicad_sch

Power



File: power.kicad_sch

In_Out



File: in_out_puts.kicad_sch

By default, everything is configured for the Nerf Laser Ops Deltaburst Blaster. By cutting off the solder jumpers and re-soldering, the board can also be configured for the Nerf Laser Ops Alphapoint Blaster. In addition, there are some new features that are optional, such as the camera(OV5060), an Accelorometer or an OLED display (SSD1306).

made by Nico Hütter

Sheet: /

File: LaserOps_PCB.kicad_sch

Title: Nerf-ATG

Size: A4

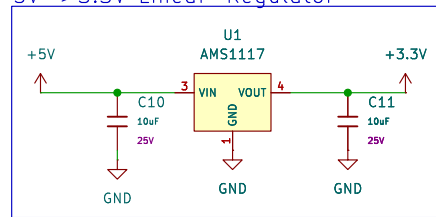
Date: 2025-04-13

Rev: 1.0

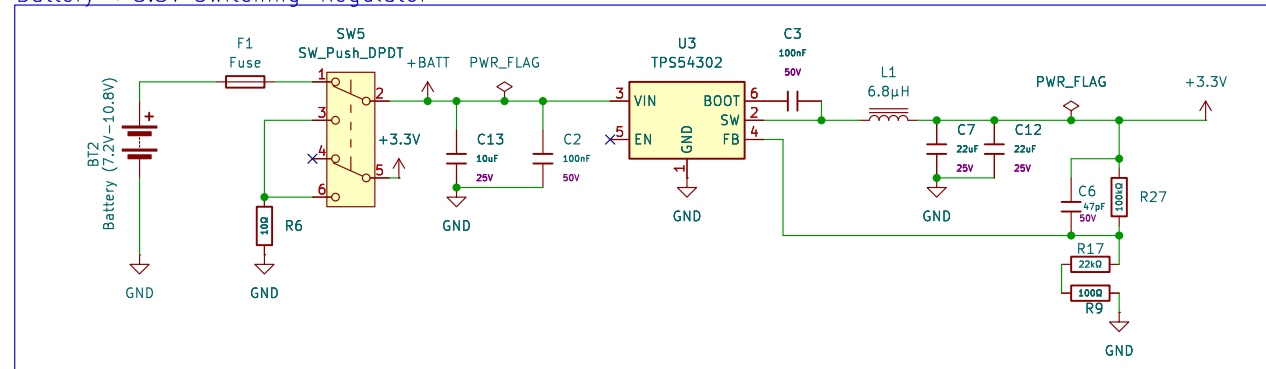
KiCad E.D.A. 9.0.0

Id: 1/4

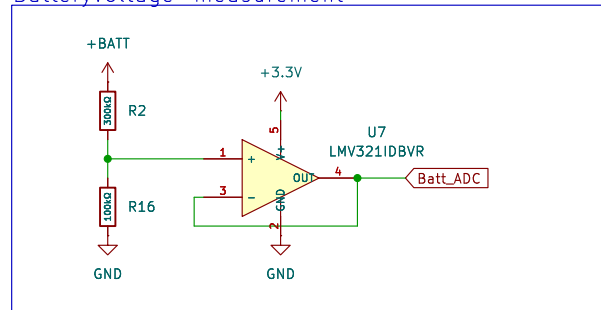
5V->3.3V Linear-Regulator



Battery->3.3V Switching-Regulator



Batteryvoltage-measurement



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Rev: 1.0

Id: 2/4

The diagram shows the wiring for the Hit Lamp. A 3.3V supply is connected to a 220Ω resistor (R5), which is in series with the Hit Lamp. The Hit Lamp is connected to pin 5 of the J3 Dome connector. The IR_Signal input is connected to pin 3 of the J3 Dome connector. The J3 Dome connector is also connected to a GND pin. A JP1 Indoor/Outdoor switch is connected between pins 1 and 2 of the J3 Dome connector.

Connectable devices:

- Displays:
 - Deltaburst display
 - SSD1306
- LED-boards:
 - Alphapoint status-LED

Conn_01x07

J8

7 IO35

6 IO35

5 +3.3V

4 IO36

3

2 IO37

1 GND

JP6 Jumper

3 IO35

2 IO35

1 SDA

JP7 Jumper

3 IO43

2 IO43

1 SCL

SD_MODE to 3.3V over a big resistor for mixed audio output.
GAIN_SLOT to GND for 120db gain.

Camera_Interface1
Conn_02x08_Counter_Clockwise

+5V

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

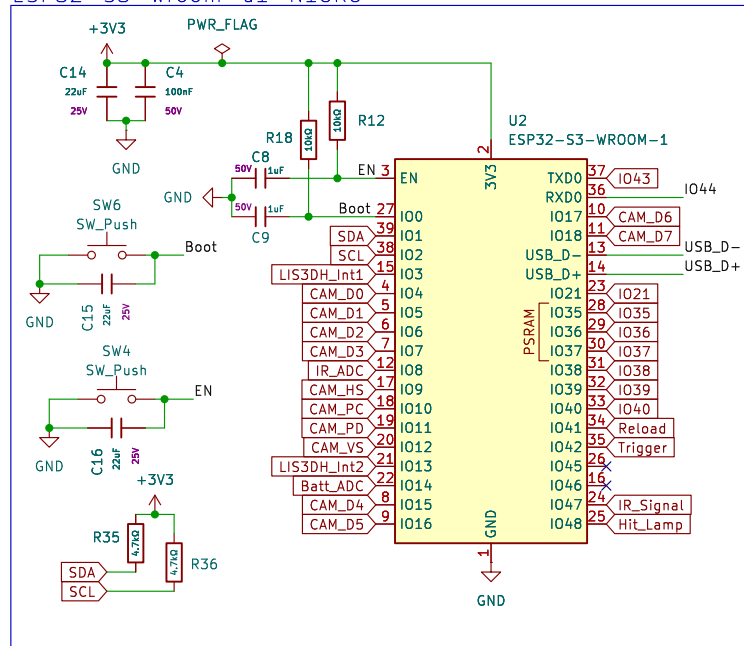
SCL CAM_VS CAM_PC CAM_D7 CAM_D5 CAM_D3 CAM_D1 CAM_D1 CAM_PD CAM_D2 CAM_D0 CAM_D4 CAM_D6 CAM_HS SDA GND

Interface for Adafruit
OV5640 Camera Breakouts

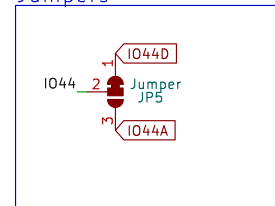
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Id: 3/4

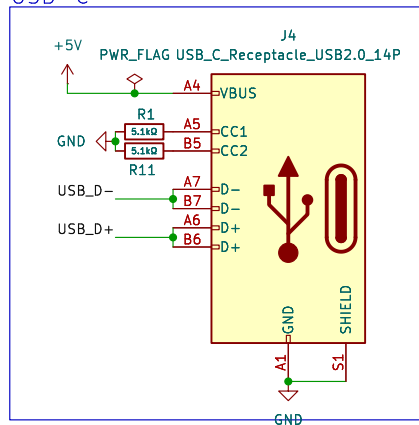
ESP32-S3-wroom-u1-N16R8



Jumpers



USB-C



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Rev: 1.0
Id: 4/4